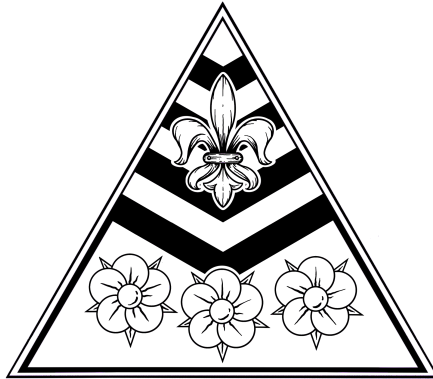
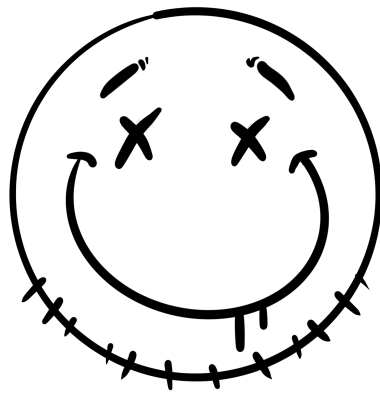




Lab 13: Log Analysis with Splunk

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1 Standing Up Splunk

The following section contains the steps taken to ensure that Splunk was installed and running successfully. Proofs are given in screenshot and written formats.

1.1 Running Splunk

```
csec3755@ubuntu:~$ docker run --name splunk --hostname splunk -p 8000:8000 -p 9997:9997 -e "SPLUNK_PASSWORD=ChangeMe123!" -e "SPLUNK_START_ARGS=--accept-license" -e "SPLUNK_GENERAL_TERMS=--accept-sgt-current-at-splunk-com" -d splunk/splunk:latest
```

Figure 1: Screenshot of `docker run` command ran on Ubuntu.

```
NAME
38f7e938190d splunk/splunk:latest "/sbin/entrypoint.sh..." 7 minutes ago Up 7 minutes (healthy) 8065/tcp, 8088-8089/tcp, 8191/tcp, 0.0.0.0:8000->8000/tcp, [::]:8000->8000/tcp, 0.0.0.0:9997->9997/tcp, [::]:9997->9997/tcp, 9887/tcp
splunk
8e2840b5c23f ghcr.io/msu-denver/medtech-patient-portal:latest "python app.py" 3 weeks ago Up About an hour 0.0.0.0:80->5000/tcp, [::]:80->5000/tcp
```

Figure 2: Screenshot of `docker ps` results showing the Splunk container is healthy.

1.2 Ubuntu's IP

- 127.0.0.1

1.3 Setting Sysmon Index in Splunk

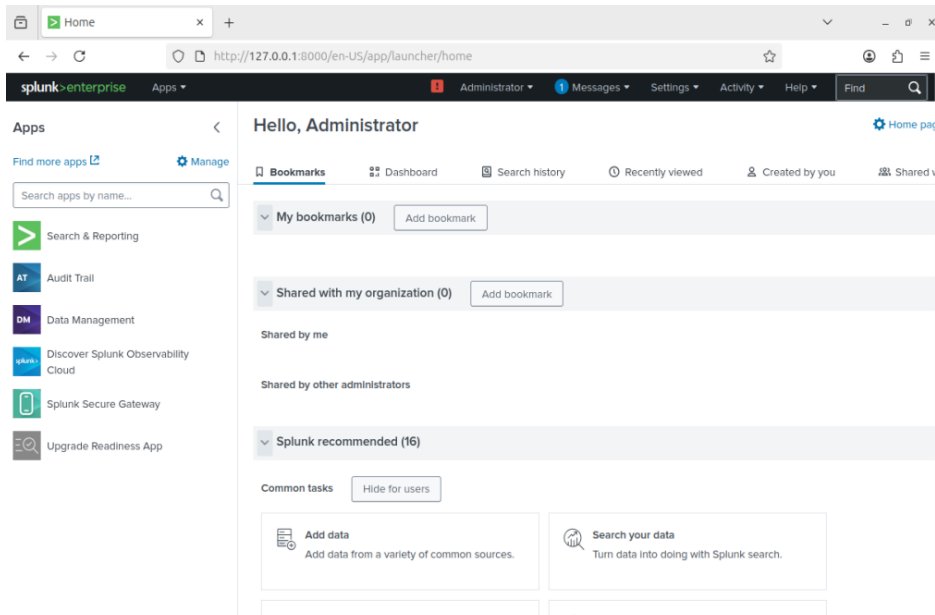


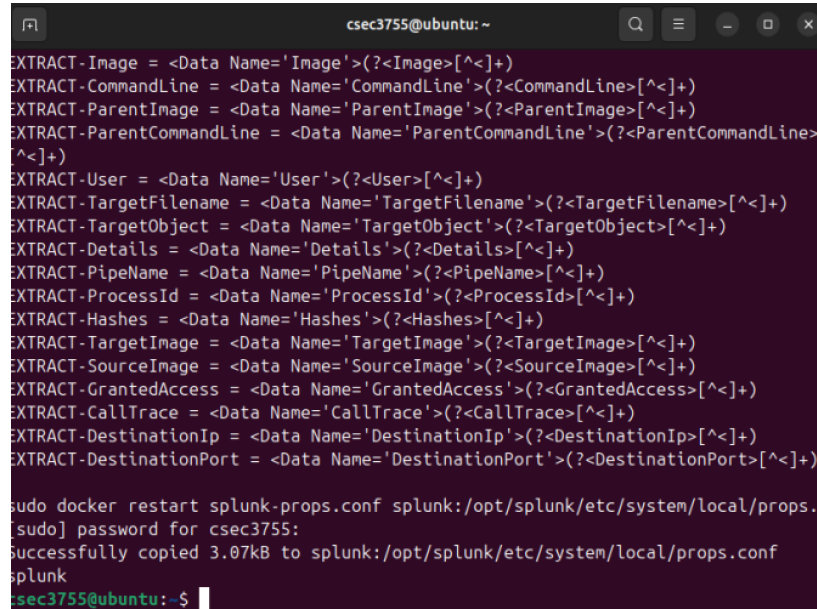
Figure 3: Screenshot of Splunk GUI homepage accessed as Administrator.

Name	Status	Type	App	Current Size	Max Size	Event Count	Earliest Event	Latest Event	Home Path	Frozen Path
_telemetry	Active	Events	system	1 MB	488.28 GB	0	–	–	\$SPLUNK_DB/_telemetry/db	–
_thefishbucket	Active	Events	system	1 MB	488.28 GB	0	–	–	\$SPLUNK_DB/fishbucket/db	–
history	Active	Events	system	1 MB	488.28 GB	0	–	–	\$SPLUNK_DB/historydb/db	–
main	Active	Events	system	1 MB	488.28 GB	0	–	–	\$SPLUNK_DB/defaultdb/db	–
splunklogger	Inactive	Events	system	0 B	488.28 GB	0	–	–	\$SPLUNK_DB/splunklogger/db	–
summary	Active	Events	system	1 MB	488.28 GB	0	–	–	\$SPLUNK_DB/summarydb/db	–
sysmon	Active	Events	search	1 MB	500 GB	0	–	–	\$SPLUNK_DB/sysmon/db	–

Figure 4: Screenshot of updated Splunk GUI index page showing 'sysmon' index.

Why a sysmon index? Creating a dedicated Sysmon index keeps your endpoint telemetry isolated from other log sources, making your SPL queries faster and more precise since you can scope searches to `index=sysmon` instead of sifting through everything in main.

1.4 Sysmon field-extractions (props.conf)



```

csec3755@ubuntu: ~
EXTRACT-Image = <Data Name='Image'>(?!<Image>[^\<]+)
EXTRACT-CommandLine = <Data Name='CommandLine'>(?!<CommandLine>[^\<]+)
EXTRACT-ParentImage = <Data Name='ParentImage'>(?!<ParentImage>[^\<]+)
EXTRACT-ParentCommandLine = <Data Name='ParentCommandLine'>(?!<ParentCommandLine>[^\<]+)
EXTRACT-User = <Data Name='User'>(?!<User>[^\<]+)
EXTRACT-TargetFilename = <Data Name='TargetFilename'>(?!<TargetFilename>[^\<]+)
EXTRACT-TargetObject = <Data Name='TargetObject'>(?!<TargetObject>[^\<]+)
EXTRACT-Details = <Data Name='Details'>(?!<Details>[^\<]+)
EXTRACT-PipeName = <Data Name='PipeName'>(?!<PipeName>[^\<]+)
EXTRACT-ProcessId = <Data Name='ProcessId'>(?!<ProcessId>[^\<]+)
EXTRACT-Hashes = <Data Name='Hashes'>(?!<Hashes>[^\<]+)
EXTRACT-TargetImage = <Data Name='TargetImage'>(?!<TargetImage>[^\<]+)
EXTRACT-SourceImage = <Data Name='SourceImage'>(?!<SourceImage>[^\<]+)
EXTRACT-GrantedAccess = <Data Name='GrantedAccess'>(?!<GrantedAccess>[^\<]+)
EXTRACT-CallTrace = <Data Name='CallTrace'>(?!<CallTrace>[^\<]+)
EXTRACT-DestinationIp = <Data Name='DestinationIp'>(?!<DestinationIp>[^\<]+)
EXTRACT-DestinationPort = <Data Name='DestinationPort'>(?!<DestinationPort>[^\<]+)

sudo docker restart splunk-props.conf splunk:/opt/splunk/etc/system/local/props.conf
[sudo] password for csec3755:
Successfully copied 3.07kB to splunk:/opt/splunk/etc/system/local/props.conf
splunk
csec3755@ubuntu:~$

```

Figure 5: Screenshot of confirmation of successful `props.conf` extraction.

2 Forwarder Setup

3 ART & SPL

4 The Dashboard

++report written with (minimal) help from Claude (Sonnet 4.6)++